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Audience: LEWG

***fiber_handle* - fibers without scheduler**

Revision History	1
abstract	1
control transfer mechanism	2
std::fiber_handle as a first-class object	3
encapsulating the stack	3
invalidation at resumption	4
problem: avoiding non-const global variables and undefined behaviour	4
solution: avoiding non-const global variables and undefined behaviour	5
inject function into suspended fiber	10
passing data between fibers	11
termination	12
exceptions	13
stack destruction	13
std::fiber_handle as building block for higher-level frameworks	13
interaction with STL algorithms	15
possible implementation strategies	15
fiber switch on architectures with register window	17
how fast is a fiber switch	17
interaction with accelerators	17
multi-threading environment	17
acknowledgment	18
API	19
references	24

Revision History

This document supersedes P0876R2.

Changes since P0876R2.

- rename `fiber_context` to `fiber_handle`
- remove `StackAllocator` constructor and related material
- rename `resume_other_thread()` to `resume_from_any_thread()`
- rename `resume_other_thread_with()` to `resume_from_any_thread_with()`
- replace `uses_system_stack()` with `can_resume_from_any_thread()` (with inverted return sense)
- replace `previous_thread()` with `can_resume()` (with bool return)

abstract

This paper addresses concerns, questions and suggestions from the past meetings. The proposed API supersedes the former proposals N3985,⁵ P0099R1,⁷ P0534R3,⁸ P0876R0⁹ and P0876R2.¹⁰

Because of name clashes with *coroutine* from coroutine TS, *execution context* from executor proposals and *continuation* used in the context of `future::then()`, the committee has indicated that *fiber* is preferable. However, given the foundational, low-level nature of this proposal, we choose *fiber_handle*, leaving the term *fiber* for a higher-level facility built on top of this one.

A previous revision of this proposal suggested *fiber_context*, but SG1 felt that use of the term “context” posed potential confusion. The name *basic_fiber* has also been suggested. Consider that the name is not yet final.

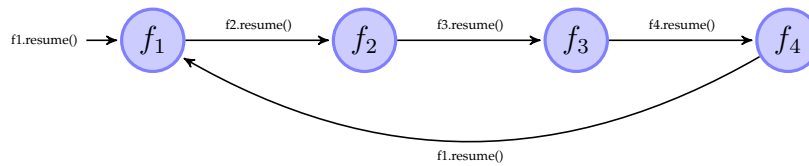
The minimal API enables stackful context switching **without** the need for a **scheduler**. The API is suitable to act as building-block for high-level constructs such as stackful coroutines as well as cooperative multitasking (aka user-land/green threads that incorporate a **scheduling facility**).

Informally within this proposal, the term *fiber* is used to denote the lightweight thread of execution launched and represented by the first-class object `std::fiber_handle`.

control transfer mechanism

According to the literature,⁴ coroutine-like control-transfer operations can be distinguished into the concepts of *symmetric* and *asymmetric* operations.

symmetric fiber A symmetric fiber provides a single control-transfer operation. This single operation requires that the control is passed explicitly between the fibers.



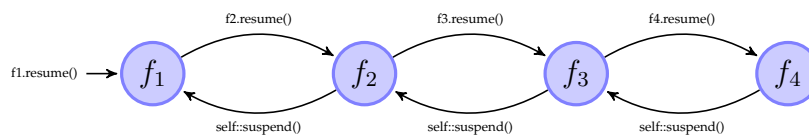
```

1  fiber_handle* pf1;
2  fiber_handle f4{[&pf1]{
3    pf1->resume();
4  }};
5  fiber_handle f3{[&f4]{
6    f4.resume();
7  }};
8  fiber_handle f2{[&f3]{
9    f3.resume();
10 }};
11 fiber_handle f1{[&f2]{
12   f2.resume();
13 }};
14 pf1=&f1;
15 f1.resume();
  
```

In the pseudo-code example above, a chain of fibers is created.

Control is transferred to fiber `f1` at line 15 and the lambda passed to constructor of `f1` is entered. Control is transferred from fiber `f1` to `f2` at line 12 and from `f2` to `f3` (line 9) and so on. Fiber `f4` itself transfers control directly back to fiber `f1` at line 3.

asymmetric fiber Two control-transfer operations are part of asymmetric fiber's interface: one operation for resuming (`resume()`) and one for suspending (`suspend()`) the fiber. The suspending operation returns control back to the calling fiber.



```

1  fiber_handle f4{[] {
2    self::suspend();
3  }};
4  fiber_handle f3{[&f4]{
5    f4.resume();
6    self::suspend();
7  }};
  
```

```

8  fiber_handle f2{[&f3]{
9      f3.resume();
10     self::suspend();
11 }};
12 fiber_handle f1{[&f2]{
13     f2.resume();
14     self::suspend();
15 }};
16 f1.resume();

```

In the pseudo code above execution control is transferred to fiber `f1` at line 16. Fiber `f1` resumes fiber `f2` at line 13 and so on. At line 2 fiber `f4` calls its suspend operation `self::suspend()`. Fiber `f4` is suspended and `f3` resumed. Inside the lambda, `f3` returns from `f4.resume()` and calls `self::suspend()` (line 6). Fiber `f3` gets suspended while `f2` will be resumed and so on ...

The asymmetric version needs **N-1 more** fiber switches than the variant using symmetric fibers.

While asymmetric fibers establish a caller-callee relationship (strongly coupled), symmetric fibers operate as siblings (loosely coupled).

Symmetric fibers represent independent units of execution, making symmetric fibers a suitable mechanism for concurrent programming. Additionally, constructs that produce sequences of values (*generators*) are easily constructed out of two symmetric fibers (one represents the caller, the other the callee).

Asymmetric fibers incorporate additional fiber switches as shown in the pseudo code above. It is obvious that for a broad range of use cases, asymmetric fibers are less efficient than their symmetric counterparts.

Additionally, the calling fiber must be kept alive until the called fiber terminates. Otherwise the call of `suspend()` will be undefined behaviour (where to transfer execution control to?).

Symmetric fibers are more efficient, have fewer restrictions (no caller-callee relationship) and can be used to create a wider set of applications (generators, cooperative multitasking, backtracking ...).

`std::fiber_handle` as a first-class object

Because the symmetric control-transfer operation requires explicitly passing control between fibers, fibers must be expressed as *first-class objects*.

Fibers exposed as first-class objects can be passed to and returned from functions, assigned to variables or stored into containers. With fibers as first-class objects, a program can **explicitly control the flow of execution** by suspending and resuming fibers, enabling control to pass into a function at exactly the point where it previously suspended.

Symmetric control-transfer operations require fibers to be first-class objects. First-class objects can be returned from functions, assigned to variables or stored into containers.

encapsulating the stack

Each fiber is associated with a stack and is responsible for managing the lifespan of its stack (allocation at construction, deallocation when fiber terminates). The RAII-pattern* should apply.

Copying a `std::fiber_handle` must not be permitted!

If a `std::fiber_handle` were copyable, then its stack with all the objects allocated on it must be copied too. That presents two implementation choices.

- One approach would be to capture sufficient metadata to permit object-by-object copying of stack contents. That would require dramatically more runtime information than is presently available – and would take considerably more overhead than a coder might expect. Naturally, any one move-only object on the stack would prohibit copying the entire stack.

*resource acquisition is initialisation

- The other approach would be a bitwise copy of the memory occupied by the stack. That would force undefined behaviour if any stack objects were RAII-classes (managing a resource via RAII pattern). When the first of the fiber copies terminates (unwinds its stack), the RAII class destructors will release their managed resources. When the second copy terminates, the same destructors will try to doubly-release the same resources, leading to undefined behavior.

A fiber API must:

- encapsulate the stack
- manage lifespan of an explicitly-allocated stack: the stack gets deallocated when `std::fiber_handle` goes out of scope
- prevent accidentally copying the stack

Class `std::fiber_handle` must be *moveable-only*.

invalidation at resumption

The framework must prevent the resumption of an already running or terminated (computation has finished) fiber. Resuming an already running fiber will cause overwriting and corrupting the stack frames (note, the stack is not copyable). Resuming a terminated fiber will cause undefined behaviour because the stack might already be unwound (objects allocated on the stack were destroyed or the memory used as stack was already deallocated).

As a consequence each call of `resume()` will invalidate the `std::fiber_handle` instance, i.e. no valid instance of `std::fiber_handle` represents the currently-running fiber.

Whether a `std::fiber_handle` is valid or not can be tested with member function `operator bool()`.

To make this more explicit, functions `resume()`, `resume_with()`, `resume_from_any_thread()` and `resume_from_any_thread_with()` are rvalue-reference qualified.

The essential points:

- regardless of the number of `std::fiber_handle` declarations, exactly one `std::fiber_handle` instance represents each suspended fiber
- no `std::fiber_handle` instance represents the currently-running fiber

Section **solution: avoiding non-const global variables and undefined behaviour** describes how an instance of `std::fiber_handle` is synthesized from the active fiber that suspends.

A fiber API must:

- prevent accidentally resuming a running fiber
- prevent accidentally resuming a terminated (computation has finished) fiber
- `resume()`, `resume_with()`, `resume_from_any_thread()` and `resume_from_any_thread_with()` are rvalue-reference qualified to bind on rvalues only

problem: avoiding non-const global variables and undefined behaviour

According to C++ *core guidelines*,¹¹ non-const global variables should be avoided: they hide dependencies and make the dependencies subject to unpredictable changes.

Global variables can be changed by assigning them indirectly using a pointer or by a function call. As a consequence, the compiler can't cache the value of a global variable in a register, degrading performance (unnecessary loads and stores to global memory especially in performance critical loops).

Accessing a register is one to three orders of magnitude faster than accessing memory (depending on whether the cache line is in cache and not invalidated by another core; and depending on whether the page is in the TLB).

The order of initialisation (and thus destruction) of static global variables is not defined, introducing additional problems with static global variables.

A library designed to be used as building block by other higher-level frameworks should avoid introducing global variables. If this API were specified in terms of internal global variables, no higher level layer could undo that: it would be stuck with the global variables.

switch back to *main()* by returning Switching back to `main()` by returning from the fiber function has two drawbacks: it requires an internal global variable pointing to the suspended `main()` and restricts the valid use cases.

```
int main() {
    fiber_handle f{[]{
        ...
        // switch to 'main()' only by returning
    }};
    f.resume(); // resume 'f'
    return 0;
}
```

For instance the generator pattern is impossible because the only way for a fiber to transfer execution control back to `main()` is to terminate. But this means that no way exists to transfer data (sequence of values) back and forth between a fiber and `main()`.

Switching to `main()` only by returning is impractical because it limits the applicability of fibers and requires an internal global variable pointing to `main()`.

static member function returns active `std::fiber_handle` P0099R0⁶ introduced a static member function (`execution_context::current()`) that returned an instance of the active fiber. This allows passing the active fiber `m` (for instance representing `main()`) into the fiber `l` via lambda capture. This mechanism enables switching back and forth between the fiber and `main()`, enabling a rich set of applications (for instance generators).

```
int main() {
    int a;
    fiber_handle m=fiber_handle::current(); // get active fiber
    fiber_handle f{[&]{
        a=0;
        int b=1;
        for(;;) {
            m=m.resume(); // switch to 'main()'
            int next=a+b;
            a=b;
            b=next;
        }
    }};
    for(int j=0; j<10; ++j) {
        f=f.resume(); // resume 'f'
        std::cout << a << " ";
    }
    return 0;
}
```

But this solution requires an internal global variable pointing to the active fiber and some kind of reference counting. Reference counting is needed because `fiber_handle::current()` necessarily requires multiple instances of `std::fiber_handle` for the active fiber. Only when the last reference goes out of scope can the fiber be destroyed and its stack deallocated.

```
fiber_handle f1=fiber_handle::current();
fiber_handle f2=fiber_handle::current();
assert(f1==f2); // f1 and f2 point to the same (active) fiber
```

Additionally a static member function returning an instance of the active fiber would violate the protection requirements of sections **encapsulating the stack** and **invalidation at resumption**. For instance you could accidentally attempt to resume the active fiber by invoking `resume()`.

```
fiber_handle m=fiber_handle::current();
m.resume(); // tries to resume active fiber == UB
```

A static member function returning the active fiber requires a reference counted global variable and does not prevent accidentally attempting to resume the active fiber.

solution: avoiding non-const global variables and undefined behaviour

The *avoid non-const global variables* guideline has an important impact on the design of the `std::fiber_handle` API!

synthesizing the suspended fiber The problem of global variables or the need for a static member function returning the active fiber can be avoided by **synthesizing the suspended fiber** and passing it into the resumed fiber (as parameter when the fiber is first started, or returned from `resume()`).

```
1 void foo() {
2     fiber_handle f{[] (fiber_handle&& m) {
3         m=std::move(m).resume(); // switch to 'foo()'
4         m=std::move(m).resume(); // switch to 'foo()'
5         ...
6     }};
7     f=std::move(f).resume(); // start 'f'
8     f=std::move(f).resume(); // resume 'f'
9 }
```

In the pseudo-code above the fiber `f` is started by invoking its member function `resume()` at line 7. This operation suspends `foo`, invalidates instance `f` and synthesizes a new `std::fiber_handle m` that is passed as parameter to the lambda of `f` (line 2).

Invoking `m.resume()` (line 3) suspends the lambda, invalidates `m` and synthesizes a `std::fiber_handle` that is returned by `f.resume()` at line 7. The synthesized `std::fiber_handle` is assigned to `f`. Instance `f` now represents the suspended fiber running the lambda (suspended at line 3). Control is transferred from line 3 (lambda) to line 7 (`foo()`). Call `f.resume()` at line 8 invalidates `f` and suspends `foo()` again. A `std::fiber_handle` representing the suspended `foo()` is synthesized, returned from `m.resume()` and assigned to `m` at line 3. Control is transferred back to the lambda and instance `m` represents the suspended `foo()`.

Function `foo()` is resumed at line 4 by executing `m.resume()` so that control returns at line 8 and so on ...

Class `symmetric_coroutine<>::yield_type` from N3985⁵ is **not** equivalent to the synthesized `std::fiber_handle`. `symmetric_coroutine<>::yield_type` does not represent the suspended context, instead it is a special representation of the same coroutine. Thus `main()` or the current thread's *entry-function* can **not** be represented by `yield_type` (see next section **representing `main()` and thread's entry-function as fiber**).

Because `symmetric_coroutine<>::yield_type()` yields back to the starting point, i.e. invocation of `symmetric_coroutine<>::call_type::operator()()`, both instances (`call_type` as well as `yield_type`) must be preserved. Additionally the caller must be kept alive until the called coroutine terminates or UB happens at resumption.

This API is specified in terms of passing the suspended `std::fiber_handle`. A higher level layer can hide that by using private variables.

representing `main()` and thread's entry-function as fiber As shown in the previous section a synthesized fiber is created and passed into the resumed fiber as an instance of `std::fiber_handle`.

```
int main() {
    fiber_handle f{[] (fiber_handle&& m) {
        m=std::move(m).resume(); // switch to 'main()'
        ...
    }};
    f=std::move(f).resume(); // resume 'f'
    return 0;
}
```

The mechanism presented in this proposal describes switching between stacks: each fiber has its own stack. The stacks of `main()` and explicitly-launched threads are not excluded; these can be used as targets too.

Thus every program can be considered to consist of fibers – some created by the OS (`main()` stack; each thread's initial stack) and some created explicitly by the code.

This is a nice feature because it allows (the stacks of) `main()` and each thread's *entry-function* to be represented as fibers. A `std::fiber_handle` representing `main()` or a thread's *entry-function* can be handled like an explicitly created `std::fiber_handle`: it can be passed to and returned from functions or stored in a container. When called on such an instance, `can_resume_from_any_thread()` indicates whether it is valid to call `resume_from_any_thread()` or

`resume_from_any_thread_with()` on that instance.

In the code snippet above the suspended `main()` is represented by instance `m` and could be stored in containers or managed just like `f` by a scheduling algorithm.

The proposed fiber API allows representing and handling `main()` and the current thread's *entry-function* by an instance of `std::fiber_handle` in the same way as explicitly created fibers.

fiber returns (terminates) When a fiber returns (terminates), what should happen next? Which fiber should be resumed next? The only way to avoid internal global variables that point to `main()` is to explicitly return a valid `std::fiber_handle` instance that will be resumed after the active fiber terminates.

```
1 int main(){
2     fiber_handle f{[] (fiber_handle&& m){
3         return std::move(m); // resume 'main()' by returning 'm'
4     }};
5     std::move(f).resume(); // resume 'f'
6     return 0;
7 }
```

In line 5 the fiber is started by invoking `resume()` on instance `f`. `main()` is suspended and an instance of type `fiber_handle` is synthesized and passed as parameter `m` to the lambda at line 2. The fiber terminates by returning `m`. Control is transferred to `main()` (returning from `f.resume()` at line 5) while fiber `f` is destroyed.

In a more advanced example another `std::fiber_handle` is used as return value instead of the passed in synthesized fiber.

```
1 int main(){
2     fiber_handle m;
3     fiber_handle f1{[&] (fiber_handle&& f){
4         std::cout << "f1: entered first time" << std::endl;
5         assert(!f);
6         return std::move(m); // resume (main-)fiber that has started 'f2'
7     }};
8     fiber_handle f2{[&] (fiber_handle&& f){
9         std::cout << "f2: entered first time" << std::endl;
10        m=std::move(f); // preserve 'f' (== suspended main())
11        return std::move(f1);
12    }};
13    std::move(f2).resume();
14    std::cout << "main: done" << std::endl;
15    return 0;
16 }
17
18 output:
19 f2: entered first time
20 f1: entered first time
21 main: done
```

At line 13 fiber `f2` is resumed and the lambda is entered at line 8. The synthesized `std::fiber_handle f` (representing suspended `main()`) is passed as a parameter `f` and stored in `m` (captured by the lambda) at line 10. This is necessary in order to prevent destructing `f` when the lambda returns. Fiber `f2` uses `f1`, that was also captured by the lambda, as return value. Fiber `f2` terminates while fiber `f1` is resumed (entered the first time). The synthesized `std::fiber_handle f` passed into the lambda at line 3 represents the terminated fiber `f2` (e.g. the calling fiber). Thus instance `f` is invalid as the assert statement verifies at line 5. Fiber `f1` uses the captured `std::fiber_handle m` as return value (line 6). Control is returned to `main()`, returning from `f2.resume()` at line 13.

The function passed to `std::fiber_handle`'s constructor must have signature '`fiber_handle(fiber_handle&&)`'. Using `std::fiber_handle` as the return value from such a function avoids global variables.

returning synthesized `std::fiber_handle` instance from `resume()` An instance of `std::fiber_handle` remains invalid after return from `resume()`, `resume_with()`, `resume_from_any_thread()` or `resume_from_any_thread_with()` – the synthesized fiber is returned, instead of implicitly updating the `std::fiber_handle` instance on which `resume()` was called.

If the `std::fiber_handle` object were implicitly updated, the fiber would change its identity because each fiber is associated with a stack. Each stack contains a chain of function calls (call stack). If this association were implicitly modified, unexpected behaviour happens.

The example below demonstrates the problem:

```

1  int main() {
2      fiber_handle f1, f2, f3;
3      f3=fiber_handle{ [&] (fiber_handle&& f)->fiber_handle{
4          f2=std::move(f);
5          for(;;) {
6              std::cout << "f3 ";
7              std::move(f1).resume();
8          }
9          return {};
10     } };
11     f2=fiber_handle{ [&] (fiber_handle&& f)->fiber_handle{
12         f1=std::move(f);
13         for(;;) {
14             std::cout << "f2 ";
15             std::move(f3).resume();
16         }
17         return {};
18     } };
19     f1=fiber_handle{ [&] (fiber_handle&& /*main*/) ->fiber_handle{
20         for(;;) {
21             std::cout << "f1 ";
22             std::move(f2).resume();
23         }
24         return {};
25     } };
26     std::move(f1).resume();
27     return 0;
28 }
29
30 output:
31 f1 f2 f3 f1 f3 f1 f3 f1 f3 ...

```

In this pseudo-code the `std::fiber_handle` object is implicitly updated.

The example creates a circle of fibers: each fiber prints its name and resumes the next fiber (`f1 -> f2 -> f3 -> f1 -> ...`).

Fiber `f1` is started at line 26. The synthesized `std::fiber_handle` `main` passed to the resumed fiber is not used (control flow cycles through the three fibers).^{*} The for-loop prints the name `f1` and resumes fiber `f2`. Inside `f2`'s for-loop the name is printed and `f3` is resumed. Fiber `f3` resumes fiber `f1` at line 7. Inside `f1` control returns from `f2.resume()`. `f1` loops, prints out the name and invokes `f2.resume()`. But this time fiber `f3` instead of `f2` is resumed. This is caused by the fact the instance `f2` gets the synthesized `std::fiber_handle` of `f3` implicitly assigned. Remember that at line 7 fiber `f3` gets suspended while `f1` is resumed through `f1.resume()`.

This problem can be solved by returning the synthesized `std::fiber_handle` from `resume()`, `resume_with()`, `resume_from_any_thread()` or `resume_from_any_thread_with()`.

```

int main() {
    fiber_handle f1, f2, f3;
    f3=fiber_handle{ [&] (fiber_handle&& f)->fiber_handle{
        f2=std::move(f);
        for(;;) {
            std::cout << "f3 ";
            f2=std::move(f1).resume();
        }
    } };

```

^{*}The operating-system stack provided for `main()` or the current thread's *entry-function* is not destroyed when the corresponding `std::fiber_handle` instance is destroyed.

```

        return {};
    }
};
f2=fiber_handle{[&](fiber_handle&& f)->fiber_handle{
    f1=std::move(f);
    for(;;){
        std::cout << "f2 ";
        f1=std::move(f3).resume();
    }
    return {};
}};
f1=fiber_handle{[&](fiber_handle&& /*main*/)->fiber_handle{
    for(;;){
        std::cout << "f1 ";
        f3=std::move(f2).resume();
    }
    return {};
}};
std::move(f1).resume();
return 0;
}

```

output:

f1 f2 f3 f1 f2 f3 f1 f2 f3 ...

In the example above the synthesized `std::fiber_handle` returned by `resume()` is move-assigned to the invoking `std::fiber_handle` instance (that has resumed the current fiber).

The synthesized `std::fiber_handle` must be returned from `resume()`, `resume_with()`, `resume_from_any_thread()` and `resume_from_any_thread_with()` in order to prevent changing the identity of the fiber.

If the overall control flow isn't known, member functions `resume_with()` or `resume_from_any_thread_with()` (see section [inject function into suspended fiber](#)) can be used to assign the synthesized `std::fiber_handle` to the correct `std::fiber_handle` instance (the caller).

```

class filament{
private:
    fiber_handle      f_;

public:
    ...
    void resume_next( filament& fila){
        std::move(fila.f_).resume_with([this](fiber_handle&& f)->fiber_handle{
            f_=std::move(f);
            return {};
        })
    }
};

```

Picture a higher-level framework in which every fiber can find its associated `filament` instance, as well as others. Every context switch must be mediated by passing *the target* `filament` instance to *the running fiber's* `resume_next()`.

Running fiber A has an associated `filament` instance `filamentA`, whose `std::fiber_handle f_` is invalid – because fiber A is running.

Desiring to switch to suspended fiber B (with associated `filament` `filamentB`), running fiber A calls `filamentA.resume_next(filamentB)`.

`resume_next()` calls `filamentB.f_.resume_with(<lambda>)`. This invalidates `filamentB.f_` – because fiber B is now running.

The lambda binds `&filamentA` as `this`. Running on fiber B, it receives a `std::fiber_handle` instance representing the newly-suspended fiber A as its parameter `f`. It moves that `std::fiber_handle` instance to `filamentA.f_`.

The lambda then returns a default-constructed (therefore invalid) `std::fiber_handle` instance. That invalid instance is returned by the previously-suspended `resume_with()` call in `filamentB.resume_next()` – which is fine because `resume_next()` drops it on the floor anyway.

Thus, the running fiber’s associated `filament::f_` is always invalid, whereas the `filament` associated with each suspended fiber is continually updated with the `std::fiber_handle` instance representing that fiber.*

It is not necessary to know the overall control flow. It is sufficient to pass a reference/pointer of the caller (fiber that gets suspended) to the resumed fiber that move-assigns the synthesized `std::fiber_handle` to caller (updating the instance).

inject function into suspended fiber

Sometimes it is useful to inject a new function (for instance, to throw an exception or assign the synthesized fiber to the caller as described in [returning synthesized std::fiber_handle instance from resume\(\)](#)) into a suspended fiber. For this purpose `resume_with(Fn&& fn)` (or `resume_from_any_thread_with()`) may be called, passing the function `fn()` to execute.

```
1 fiber_handle f([](fiber_handle&& caller){
2     // ...
3     std::move(caller).resume();
4     // ...
5 });
6
7 fiber_handle fn(fiber_handle&&);
8
9 f = std::move(f).resume();
10 // ...
11 std::move(f).resume_with(fn);
```

The `resume_with()` call at line 11 injects function `fn()` into fiber `f` as if the `resume()` call at line 3 had directly called `fn()`.

Like an *entry-function* passed to `std::fiber_handle`, `fn()` must accept `std::fiber_handle&&` and return `std::fiber_handle`. The `std::fiber_handle` instance returned by `fn()` will, in turn, be returned to `f`’s lambda by the `resume()` at line 3.

Suppose that code running on the program’s main fiber calls `resume()` (line 12 below), thereby entering the first lambda shown below. This is the point at which `m` is synthesized and passed into the lambda at line 2.

Suppose further that after doing some work (line 4), the lambda calls `m.resume()`, thereby switching back to the main fiber. The lambda remains suspended in the call to `m.resume()` at line 5.

At line 18 the main fiber calls `f.resume_with()` where the passed lambda accepts `fiber_handle&&`. That new lambda is called on the fiber of the suspended lambda. It is as if the `m.resume()` call at line 8 directly called the second lambda.

The function passed to `resume_with()` has almost the same range of possibilities as any function called on the fiber represented by `f`. Its special invocation matters when control leaves it in either of two ways:

1. If it throws an exception, that exception unwinds all previous stack entries in that fiber (such as the first lambda’s) as well, back to a matching `catch` clause.[†]
2. If the function returns, the returned `std::fiber_handle` instance is returned by the suspended `m.resume()` (or `resume_with()`, or `resume_from_any_thread()`, or `resume_from_any_thread_with()`) call.

```
1 int data = 0;
2 fiber_handle f{[&data](fiber_handle&& m){
3     std::cout << "f1: entered first time: " << data << std::endl;
4     data+=1;
```

* [Boost.Fiber¹⁵](#) uses this pattern for resuming user-land threads.

[†] As stated in [exceptions](#), if there is no matching `catch` clause in that fiber, `std::terminate()` is called.

```

5      m=std::move(m).resume();
6      std::cout << "f1: entered second time: " << data << std::endl;
7      data+=1;
8      m=std::move(m).resume();
9      std::cout << "f1: entered third time: " << data << std::endl;
10     return std::move(m);
11 };
12 f=std::move(f).resume();
13 std::cout << "f1: returned first time: " << data << std::endl;
14 data+=1;
15 f=std::move(f).resume();
16 std::cout << "f1: returned second time: " << data << std::endl;
17 data+=1;
18 f=std::move(f).resume_with([&data](fiber_handle&& m){
19     std::cout << "f2: entered: " << data << std::endl;
20     data=-1;
21     return std::move(m);
22 });
23 std::cout << "f1: returned third time" << std::endl;
24
25 output:
26     f1: entered first time: 0
27     f1: returned first time: 1
28     f1: entered second time: 2
29     f1: returned second time: 3
30     f2: entered: 4
31     f1: entered third time: -1
32     f1: returned third time

```

The `f.resume_with(<lambda>)` call at line 18 passes control to the second lambda on the fiber of the first lambda.

As usual, `resume_with()` synthesizes a `std::fiber_handle` instance representing the calling fiber, passed into the lambda as `m`. This particular lambda returns `m` unchanged at line 21; thus that `m` instance is returned by the `resume()` call at line 8.

Finally, the first lambda returns at line 10 the `m` variable updated at line 8, switching back to the main fiber.

One case worth pointing out is when you call `resume_with()` (or `resume_from_any_thread_with()`) on a `std::fiber_handle` that has not yet been resumed for the first time:

```

1  fiber_handle topfunc(fiber_handle&& prev);
2  fiber_handle injected(fiber_handle&& prev);
3
4  fiber_handle f(topfunc);
5  // topfunc() has not yet been entered
6  std::move(f).resume_with(injected);

```

In this situation, `injected()` is called with a `std::fiber_handle` instance representing the caller of `resume_with()`. When `injected()` eventually returns that (or some other valid) `std::fiber_handle` instance, the returned `std::fiber_handle` instance is passed into `topfunc()` as its `prev` parameter.

Member functions `resume_with()` and `resume_from_any_thread_with()` allow you to inject a function into a suspended fiber.

passing data between fibers

Data can be transferred between two fibers via global pointer, a calling wrapper (like `std::bind`) or lambda capture.

```

1  int i=1;
2  std::fiber_handle lambda{[&i](fiber_handle&& caller){
3      std::cout << "inside lambda,i==" << i << std::endl;
4      i+=1;
5      caller=std::move(caller).resume();

```

```

6     return std::move(caller);
7 };
8 lambda=std::move(lambda).resume();
9 std::cout << "i==" << i << std::endl;
10 lambda=std::move(lambda).resume();
11
12 output:
13     inside lambda,i==1
14     i==2

```

The `resume()` call at line 8 enters the lambda and passes 1 into the new fiber. The value is incremented by one, as shown at line 4. The expression `caller.resume()` at line 5 resumes the original context (represented within the lambda by `caller`).

The call to `lambda.resume()` at line 10 resumes the lambda, returning from the `caller.resume()` call at line 5. The `std::fiber_handle` instance `caller` invalidated by the `resume()` call at line 5 is replaced with the new instance returned by that same `resume()` call.

Finally the lambda returns (the updated) `caller` at line 6, terminating its context.

Since the updated `caller` represents the fiber suspended by the call at line 10, control returns to `main()`.

However, since context `lambda` has now terminated, the updated `lambda` is invalid. Its `operator bool()` returns `false`.

Using lambda capture is the preferred way to transfer data between two fibers; global pointers or a calling wrapper (such as `std::bind`) are alternatives.

termination

There are a few different ways to terminate a given fiber without terminating the whole process, or engaging undefined behavior.

When a `std::fiber_handle` instance is constructed with an *entry-function*, its new stack is initialized with the frame of an implicit top-level function that can catch `std::unwind_exception`. `std::unwind_exception` binds a `std::fiber_handle` instance; the implicit `catch` clause returns the bound `std::fiber_handle` from that top-level function.

Therefore, any of the following will gracefully terminate a fiber:

- Cause its *entry-function* to return a valid `std::fiber_handle`.
- From within the fiber you wish to terminate, call `std::unwind_fiber()` with a valid `std::fiber_handle`. This throws a `std::unwind_exception` instance that binds the passed `std::fiber_handle`; that fiber will be resumed when the active fiber terminates.
- From within the fiber you wish to terminate, construct and throw `std::unwind_exception`, binding the `std::fiber_handle` you intend to resume next. This is what `std::unwind_fiber()` does internally.
- Call `fiber_handle::resume_with(unwind_fiber)`. This is what `~fiber_handle()` does. Since `std::unwind_fiber()` accepts a `std::fiber_handle`, and since `resume_with()` synthesizes a `std::fiber_handle` representing its caller and passes it to the subject function, this terminates the fiber referenced by the original `std::fiber_handle` instance and switches back to the caller.
- Engage `~fiber_handle()`: switch to some other fiber, which will receive a `std::fiber_handle` instance representing the current fiber. Make that other fiber destroy the received `std::fiber_handle` instance.

(However, since the operating system allocates the stack for `main()` and for a thread's *entry-function*, of course there is no implicit top-level stack frame, no implicit `catch` (`std::unwind_exception`). In a conforming implementation, returning from a thread's *entry-function* may terminate all fibers on that thread. Returning from `main()` may terminate the whole process.)

The above are all equivalent: stack variables are properly destroyed, since the stack is unwound by throwing an exception. (See [stack destruction](#).)

In an environment that forbids exceptions, every `std::fiber_handle` you launch must terminate gracefully, by returning from its top-level function. You may not call `std::unwind_fiber()`. You may not call `~fiber_handle()`,

explicitly or implicitly, on a valid `std::fiber_handle` instance.

When an explicitly-launched fiber's *entry-function* returns a valid `std::fiber_handle` instance, that fiber is terminated. Control switches to the fiber indicated by the returned `std::fiber_handle` instance. The *entry-function* may return (switch to) any reachable valid `std::fiber_handle` instance – it need not be the instance originally passed in, or an instance returned from any of the `resume()` family of methods.

Returning an invalid `std::fiber_handle` instance (`operator bool()` returns `false`) invokes undefined behavior.

Calling `resume()` means: “Please switch to the indicated fiber; I am suspending; please resume me later.”

Returning a particular `std::fiber_handle` means: “Please switch to the indicated fiber; and by the way, I am done.”

exceptions

In general, if an uncaught exception escapes from the *entry-function*, `std::terminate` is called. There is one exception: `std::unwind_exception`. The `std::fiber_handle` facility internally uses `std::unwind_exception` to clean up the stack of a suspended context being destroyed. This exception must be allowed to propagate out of an *entry-function*.

A correct *entry-function* `try/catch` block looks like this:

```
try{
    // ... body of fiber logic ...
}catch(const std::unwind_exception&){
    // do not swallow unwind_exception
    throw;
}catch(...) {
    // ... log, or whatever ...
}
```

Of course, no `try/catch` block is needed if neither *entry-function* nor anything it calls throws exceptions.

stack destruction

On construction of a `std::fiber_handle` a stack is allocated. If the *entry-function* returns, the stack will be destroyed. If the function has not yet returned and the (destructor) of the `std::fiber_handle` instance representing that context is called, the stack will be unwound and destroyed.

Consider a running fiber `f2` that destroys the `std::fiber_handle` instance representing `f1`.

`f1`'s destructor, running on `f2`, implicitly calls member-function `resume_with()`, passing `std::unwind_fiber()` as argument. Fiber `f1` will be temporarily resumed and `std::unwind_fiber()` is invoked. Function `std::unwind_fiber()` binds an instance of `std::fiber_handle` that represents `f2`, then throws exception `std::unwind_exception`, which unwinds `f1`'s stack (walking the stack and destroying automatic variables in reverse order of construction). The first frame on `f1`'s stack, the one created by `std::fiber_handle`'s constructor, catches the exception, extracts the bound `std::fiber_handle` representing `f2` and terminates `f1` by returning `f2`. Control is returned to `f2` and `f1`'s stack gets deallocated.

The stack on which `main()` is executed, as well as the stack implicitly created by `std::thread`'s constructor, is allocated by the operating system. Such stacks are recognized by `std::fiber_handle`, and are not deallocated by its destructor.

`std::fiber_handle` as building block for higher-level frameworks

A low-level API enables a rich set of higher-level frameworks that provide specific syntaxes/semantics suitable for specific domains. As an example, the following four frameworks are based on the low-level fiber switching API of [Boost.Context](#)¹³ (implements the API of this proposal).

[Boost.Coroutine2](#)¹⁴ implements **asymmetric coroutines** `coroutine<>::push_type` and `coroutine<>::pull_type`, providing a unidirectional transfer of data. These stackful coroutines are only used in pairs. When `coroutine<>::push_type` is explicitly instantiated, `coroutine<>::pull_type` is synthesized and passed as parameter into the coroutine function. In the example below, `coroutine<>::push_type` (variable `writer`) provides the resume operation, while `coroutine<>::pull_type` (variable `in`) represents the suspend operation. Inside the lambda, `in.get()` pulls strings provided by `coroutine<>::push_type`'s output iterator support.

```

struct FinalEOL{ ~FinalEOL(){ std::cout << std::endl; } };
std::vector<std::string> words{
    "peas", "porridge", "hot", "peas",
    "porridge", "cold", "peas", "porridge",
    "in", "the", "pot", "nine",
    "days", "old" };
int num=5,width=15;
boost::coroutines2::coroutine<std::string>::push_type writer{
    [&](boost::coroutines2::coroutine<std::string>::pull_type& in){
        FinalEOL eol;
        for (;;) {
            for (int i=0; i<num; ++i){
                if (!in){
                    return;
                }
                std::cout << std::setw(width) << in.get();
                in();
            }
            std::cout << std::endl;
        }
    };
std::copy(std::begin(words), std::end(words), std::begin(writer));

```

Synca¹⁹ (by Grigory Demchenko) is a small, efficient library to perform asynchronous operations in synchronous manner. The main features are a **GO-like** syntax, support for transferring execution context explicitly between different thread pools or schedulers (portals/teleports) and asynchronous network support.

```

int fibo(int v){
    if (v<2) return v;
    int v1,v2;
    Waiter()
        .go([v,&v1]{ v1=fibo(v-1); })
        .go([v,&v2]{ v2=fibo(v-2); })
        .wait();
    return v1+v2;
}

```

The code itself looks like synchronous invocations while internally it uses asynchronous scheduling.

Boost.Fiber¹⁵ implements **user-land threads** and combines fibers with schedulers (scheduler-algorithms are customization points). The API is modelled after the `std::thread`-API and contains objects like `future`, `mutex`, `condition_variable`...

```

boost::fibers::unbuffered_channel<unsigned int> chan;
boost::fibers::fiber f1{[&chan]{
    chan.push(1);
    chan.push(1);
    chan.push(2);
    chan.push(3);
    chan.push(5);
    chan.push(8);
    chan.push(12);
    chan.close();
}};
boost::fibers::fiber f2{[&chan]{
    for (unsigned int value: chan) {
        std::cout << value << " ";
    }
    std::cout << std::endl;
}};
f1.join();
f2.join();

```

Facebook's [folly::fibers](#)¹⁸ is an asynchronous C++ framework using **user-land threads** for parallelism. In contrast to [Boost.Fiber](#), [folly::fibers](#) exposes the scheduler and permits integration with various event dispatching libraries.

```
folly::EventBase ev_base;
auto& fiber_manager=folly::fibers::getFiberManager(ev_base);
folly::fibers::Baton baton;
fiber_manager.addTask([&]{
    std::cout << "task 1: start" << std::endl;
    baton.wait();
    std::cout << "task 1: after baton.wait()" << std::endl;
});
fiber_manager.addTask([&]{
    std::cout << "task 2: start" << std::endl;
    baton.post();
    std::cout << "task 2: after baton.post()" << std::endl;
});
ev_base.loop();
```

[folly::fibers](#) is used in many critical applications at Facebook for instance in [mcrouter](#)¹⁶ and some other Facebook services/libraries like ServiceRouter (routing framework for [Thrift](#)¹⁷), Node API (graph ORM API for graph databases) ...

As shown in this section a low-level API can act as building block for a rich set of high-level frameworks designed for specific application domains that require different aspects of design, semantics and syntax.

interaction with STL algorithms

In the following example STL algorithm `std::generator` and fiber `g` generate a sequence of Fibonacci numbers and store them into `std::vector v`.

```
int a;
std::fiber_handle g([&a](std::fiber_handle&& m){
    a=0;
    int b=1;
    for(;;){
        m=std::move(m).resume();
        int next=a+b;
        a=b;
        b=next;
    }
    return std::move(m);
});
std::vector<int> v(10);
std::generate(v.begin(), v.end(), [&a,&g]() mutable {
    g=std::move(g).resume();
    return a;
});
std::cout << "v: ";
for (auto i: v) {
    std::cout << i << " ";
}
std::cout << "\n";
```

output: v: 0 1 1 2 3 5 8 13 21 34

The proposed fiber API does not require modifications of the STL and can be used together with existing STL algorithms.

possible implementation strategies

This proposal does NOT seek to standardize any particular implementation or impose any specific calling convention!

Modern **micro-processors** are **register machines**; the content of processor registers represent the execution context of the program at a given point in time.

Operating systems maintain for each process all relevant data (execution context, other hardware registers etc.) in the process table. The operating system's **CPU scheduler** periodically suspends and resumes processes in order to share CPU time between multiple processes. When a process is suspended, its execution context (processor registers, instruction pointer, stack pointer, ...) is stored in the associated process table entry. On resumption, the CPU scheduler loads the execution context into the CPU and the process continues execution.

The CPU scheduler does a **full context switch**. Besides preserving the execution context (complete CPU state), the cache must be invalidated and the memory map modified.

A kernel-level context switch is several orders of magnitude slower than a context switch at user-level.³

hypothetical fiber preserving complete CPU state This strategy tries to preserve the complete CPU state, e.g. all CPU registers. This requires that the implementation identifies the concrete micro-processor type and supported processor features. For instance the x86-architecture has several flavours of extensions such as MMX, SSE1-4, AVX1-2, AVX-512. Depending on the detected processor features, implementations of certain functionality must be switched on or off. The CPU scheduler in the operating system uses such information for context switching between processes.

A fiber implementation using this strategy requires such a detection mechanism too (equivalent to `swapper/system_32()` in the Linux kernel).

Aside from the complexity of such detection mechanisms, preserving the complete CPU state for each fiber switch is expensive.

A context switch facility that preserves the complete CPU state like an operating system is possible but impractical for user-land.

fiber switch using the calling convention For `std::fiber_handle`, not all registers need be preserved because the context switch is effected by a visible function call. It need not be completely transparent like an operating-system context switch; it only needs to be as transparent as a call to any other function. The calling convention – the part of the ABI that specifies how a function's arguments and return values are passed – determines which subset of micro-processor registers must be preserved by the called subroutine.

The **calling convention**¹² of **SYSV ABI** for **x86_64** architecture determines that general purpose registers R12, R13, R14, R15, RBX and RBP must be preserved by the sub-routine - the first arguments are passed to functions via RDI, RSI, RDX, RCX, R8 and R9 and return values are stored in RAX, RDX.

So on that platform, the `resume()` implementation preserves the **general purpose registers** (R12-R15, RBX and RBP) specified by the calling convention. In addition, the **stack pointer** and **instruction pointer** are preserved and exchanged too – thus, from the point of view of calling code, `resume()` behaves like an ordinary function call.

In other words, `resume()` acts on the level of a simple function invocation – with the same performance characteristics (in terms of CPU cycles).

This technique is used in [Boost.Context](#)¹³ which acts as building block for [folly::fibers](#). The [folly::fibers](#) framework itself is the basis of many critical applications at Facebook, such as [mcrouter](#)¹⁶ and some other Facebook services/libraries like [ServiceRouter](#) (routing framework for [Thrift](#)¹⁷), [Node API](#) (graph ORM API for graph databases) ...

in-place substitution at compile time During code generation, a compiler-based implementation could inject the assembler code responsible for the fiber switch directly into each function that calls `resume()`. That would save an extra indirection (JMP + PUSH/MOV of certain registers used to invoke `resume()`).

CPU state at the stack Because each fiber must preserve CPU registers at suspension and load those registers at resumption, some storage is required.

Instead of allocating extra memory for each fiber, an implementation can use the stack by simply advancing the stack pointer at suspension and pushing the CPU registers (CPU state) onto the stack owned by the suspending fiber. When the fiber is resumed, the values are popped from the stack and loaded into the appropriate registers.

This strategy works because only a running fiber creates new stack frames (moving the stack pointer). While a fiber is suspended, it is safe to keep the CPU state on its stack.

Using the stack as storage for the CPU state has the additional advantage that `std::fiber_handle` must only contain a pointer to the stack location: its memory footprint can be that of a pointer.

Section [synthesizing the suspended fiber](#) describes how global variables are avoided by synthesizing a `std::fiber_handle` from the active fiber (execution context) and passing this synthesized `std::fiber_handle` (representing the now-suspended fiber) into the resumed fiber. Using the stack as storage makes this mechanism very easy to implement.* Inside `resume()` the code pushes the relevant CPU registers onto the stack, and from the resulting stack address creates a new `std::fiber_handle`. This instance is then passed (or returned) into the resumed fiber (see [synthesizing the suspended fiber](#)).

Using the active fiber's stack as storage for the CPU state is efficient because no additional allocations or deallocations are required.

fiber switch on architectures with register window

The implementation of fiber switch is possible – many libc implementations still provide the `ucontext`-API (`swapcontext()` and related functions)[†] for architectures using a register window (such as SPARC). The implementation of `swapcontext()` could be used as blueprint for a fiber implementation.

how fast is a fiber switch

A fiber switch takes 11 CPU cycles on a *x86_64-Linux* system[‡] using an implementation based on the strategy described in [fiber switch using the calling convention](#) (implemented in [Boost.Context](#)¹³ branch *fiber*).

interaction with accelerators

For many core devices several programming models, such as OpenACC, CUDA, OpenCL etc., have been developed targeting **host-directed** execution using an attached or integrated accelerator. The CPU executes the main program while controlling the activity of the accelerator. Accelerator devices typically provide capabilities for efficient vector processing[§]. Usually the host-directed execution uses **computation offloading** that permits executing computationally intensive work on a separate device (accelerator).¹

For instance CUDA devices use a **command buffer** to establish communication between host and device. The host puts commands (op-codes) into the command buffer and the device process them **asynchronously**.²

It is obvious that a fiber switch does **not** interact with **host-directed device-offloading**. A fiber switch works like a function call (see [fiber switch using the calling convention](#)).

multi-threading environment

Member function `can_resume_from_any_thread()` returns `false` if the stack used by the `std::fiber_handle` instance was created by the operating system (main application or thread stack), and the caller is running on a different thread. When the stack represented by the `std::fiber_handle` instance was created by `std::fiber_handle`'s constructor, `can_resume_from_any_thread()` returns `true`.[¶] You must not attempt to resume an instance representing a stack provided by the operating system on some other thread!

`can_resume()` can be called to determine whether a given `std::fiber_handle` instance might safely be resumed on a particular thread by calling `resume()` or `resume_with()`.

`std::fiber_handle` is TLS-agnostic - best practices related to TLS apply to fibers too (see P0772R0.)

There could potentially be Undefined Behavior if:

- code running on a fiber references `thread_local` variables

*The implementation of [Boost.Context](#)¹³ utilizes this technique.

[†]`ucontext` was removed from POSIX standard by POSIX.1-2008

[‡]Intel XEON E5 2620v4 2.2GHz

[§]warp on CUDA devices, wavefront on AMD GPUs, 512-bit SIMD on Intel Xeon Phi

[¶]A possible implementation could mark the first stack frame by creating a special marker (for instance `0xBADCAFFEE` etc.) at a specific offset in the first stack frame or use a special function name for the first function and walk the stack searching for these markers.

- the compiler/runtime implementation caches a pointer to `thread_local` storage on the stack
- that fiber is suspended, and
- the suspended fiber is resumed on a different thread.

The cached TLS pointer is now pointing to storage belonging to some other thread. If the original thread terminates before the new thread, the cached TLS pointer is now dangling.

For a runtime that caches TLS pointers in such fashion, an implementation of `resume_from_any_thread()` or `resume_from_any_thread_with()` could conceivably walk the suspended stack, patching cached pointers.

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API

```
class fiber_handle {
public:
    fiber_handle() noexcept;

    template<typename Fn>
    explicit fiber_handle(Fn&& fn);

    ~fiber_handle();

    fiber_handle(fiber_handle&& other) noexcept;
    fiber_handle& operator=(fiber_handle&& other) noexcept;
    fiber_handle(const fiber_handle& other) noexcept = delete;
    fiber_handle& operator=(const fiber_handle& other) noexcept = delete;

    fiber_handle resume() &&;
    template<typename Fn>
    fiber_handle resume_with(Fn&& fn) &&;
    fiber_handle resume_from_any_thread() &&;
    template<typename Fn>
    fiber_handle resume_from_any_thread_with(Fn&& fn) &&;

    bool can_resume() noexcept;
    bool can_resume_from_any_thread() noexcept;

    explicit operator bool() const noexcept;
    bool operator<(const fiber_handle& other) const noexcept;
    void swap(fiber_handle& other) noexcept;
};
```

member functions

(constructor) constructs new `fiber_handle`

<code>fiber_handle() noexcept</code>	(1)
<code>template<typename Fn> explicit fiber_handle(Fn&& fn)</code>	(2)
<code>fiber_handle(fiber_handle&& other) noexcept</code>	(3)
<code>fiber_handle(const fiber_handle& other) = delete</code>	(4)

- 1) this constructor instantiates an invalid `std::fiber_handle`. Its `operator bool()` returns `false`.
- 2) takes a callable (function, lambda, object with `operator()()`) as argument. The callable must have signature as described in [solution: avoiding non-const global variables and undefined behaviour](#).
- 3) moves underlying state to new `std::fiber_handle`
- 4) copy constructor deleted

Notes

The entry-function `fn` is *not* immediately entered. The stack and any other necessary resources are created on construction, but `fn` is not entered until `resume()`, `resume_with()`, `resume_from_any_thread()` or `resume_from_any_thread_with()` is called.

The entry-function `fn` passed to `std::fiber_handle` will be passed a synthesized `std::fiber_handle` instance representing the suspended caller of `resume()`, `resume_with()`, `resume_from_any_thread()` or `resume_from_any_thread_with()`.

(destructor) destroys a fiber

```
~fiber_handle() (1)
```

- 1) destroys a `std::fiber_handle` instance. If this instance represents a fiber of execution (`operator bool()` returns `true`), then the fiber of execution is destroyed too. Specifically, the stack is unwound by throwing `std::unwind_exception`.*

operator= moves the `std::fiber_handle` object

```
fiber_handle& operator=(fiber_handle&& other) noexcept (1)
```

```
fiber_handle& operator=(const fiber_handle& other)=delete (2)
```

- 1) assigns the state of `other` to `*this` using move semantics
2) copy assignment operator deleted

Parameters

other another `std::fiber_handle` to assign to this object

Return value

***this**

Postcondition

- 1) `other` is invalidated (`operator bool()` returns `false`)

resume() resumes a fiber

```
fiber_handle resume() && (1)
```

```
template<typename Fn>  
fiber_handle resume_with(Fn&& fn) && (2)
```

```
fiber_handle resume_from_any_thread() && (3)
```

```
template<typename Fn>  
fiber_handle resume_from_any_thread_with(Fn&& fn) && (4)
```

- 1),3) suspends the active fiber, resumes fiber `*this`
2),4) suspends the active fiber, resumes fiber `*this` but calls `fn()` in the resumed fiber (as if called by the suspended function)

Parameters

fn function injected into resumed fiber

Return value

fiber_handle the returned instance represents the fiber that has been suspended in order to resume the current fiber

Exceptions

- 1) `resume()` or `resume_with()` might throw `std::domain_error` if the current `std::thread` is not the same as the thread on which `*this` was most recently run
2) `resume()`, `resume_with()`, `resume_from_any_thread()` or `resume_from_any_thread_with()` might throw `std::unwind_exception` if, while suspended, the `std::fiber_handle` instance representing the suspended fiber is destroyed
3) `resume()`, `resume_with()`, `resume_from_any_thread()` or `resume_from_any_thread_with()` might throw *any* exception if, while suspended:
 - some other fiber calls `resume_with()` or `resume_from_any_thread_with()` to resume this suspended fiber

* In a program in which exceptions are thrown, it is prudent to code a fiber's *entry-function* with a last-ditch `catch (...)` clause: in general, exceptions must *not* leak out of the *entry-function*. However, since stack unwinding is implemented by throwing an exception, a correct *entry-function* `try` statement must also `catch (std::unwind_exception const&)` and rethrow it.

- the function `fn` passed to `resume_with()` or `resume_from_any_thread_with()` – or some function called by `fn` – throws an exception
- 4) Any exception thrown by the function `fn` passed to `resume_with()` or `resume_from_any_thread_with()`, or any function called by `fn`, is thrown in the fiber referenced by `*this` rather than in the fiber of the caller of `resume_with()` or `resume_from_any_thread_with()`.

Preconditions

- 1) `*this` represents a valid fiber (`operator bool()` returns `true`)
- 2) for `resume()` and `resume_with()`, the current `std::thread` is the same as the thread on which `*this` was most recently run
- 3) for `resume()`, `resume_with()`, `resume_from_any_thread()` and `resume_from_any_thread_with()`, if `can_resume_from_any_thread()` would return `false`, the current `std::thread` is the same as the thread represented by `*this`

Postcondition

- 1) `*this` is invalidated (`operator bool()` returns `false`)

Notes

The intent of the distinction between `resume()` and `resume_from_any_thread()`, as between `resume_with()` and `resume_from_any_thread_with()`, is both for validation and for code auditing. If an application only ever calls `resume()` and `resume_with()`, no fiber will ever be resumed on a thread other than the one on which it was initially resumed. Any attempt to do so will throw.

The intent of the names `resume_from_any_thread()` and `resume_from_any_thread_with()` is to clarify the direction in which cross-thread resumption occurs. The calling thread always directly resumes a suspended fiber: control is passed into the suspended fiber, and the currently-running fiber suspends. These method names mean that the fiber represented by `*this` will be resumed whether or not it was last resumed on the calling thread.

`resume()`, `resume_with()`, `resume_from_any_thread()` and `resume_from_any_thread_with()` preserve the execution context of the calling fiber. Those data are restored if the calling fiber is resumed.

A suspended `fiber_handle` can be destroyed. Its resources will be cleaned up at that time.

The returned `fiber_handle` indicates via `operator bool()` whether the previous active fiber has terminated (returned from *entry-function*).

Because `resume()`, `resume_with()`, `resume_from_any_thread()` and `resume_from_any_thread_with()` invalidate the instance on which they are called, *no valid `std::fiber_handle` instance ever represents the currently-running fiber*. In order to express the invalidation explicitly, these methods are rvalue-reference qualified.

When calling any of these methods, it is conventional to replace the newly-invalidated instance – the instance on which the method was called – with the new instance returned by that call. This helps to avoid subsequent inadvertent attempts to resume the old, invalidated instance.

An injected function `fn()` must accept `std::fiber_handle&&` and return `std::fiber_handle`. It will be passed a synthesized `std::fiber_handle` instance representing the suspended caller of `resume_with()` or `resume_from_any_thread_with()`. The `std::fiber_handle` instance returned by `fn()` is, in turn, used as the return value for the suspended function: `resume()`, `resume_with()`, `resume_from_any_thread()` or `resume_from_any_thread_with()`.

`can_resume_from_any_thread()` query whether the calling thread can resume the suspended `std::fiber_handle` instance by calling `resume_from_any_thread()` or `resume_from_any_thread_with()`. The implementation must return `false` if the suspended `std::fiber_handle` instance represents a fiber with a system-provided stack, and the calling thread is not that thread.

```
bool can_resume_from_any_thread() noexcept (1)
```

- 1) `fiber_handle::can_resume_from_any_thread()` returns `false` if the stack used by the fiber was provided by the operating system, and the calling thread is not that thread; otherwise `true`.

Precondition

- 1) `*this` represents a valid fiber (`operator bool()` returns `true`)

Notes

When `main()`, or the *entry-function* of a `std::thread`, or any function directly called by these, is suspended, a `std::fiber_handle` instance represents that suspended fiber. You may resume that suspended fiber *on the same thread* using any of `resume()`, `resume_with()`, `resume_from_any_thread()` or `resume_from_any_thread_with()`. Attempting to resume that suspended fiber from any other thread is Undefined Behavior.

`can_resume_from_any_thread()` returns `true` if the calling thread is the same as the thread represented by the `std::fiber_handle` instance, or if the `std::fiber_handle` instance represents a fiber explicitly created by `std::fiber_handle`'s constructor.
`can_resume_from_any_thread()` is not marked `const` because in at least one implementation, it requires an internal context switch.

`can_resume()` returns `true` if the `std::fiber_handle` instance was suspended on the same thread as the caller, or if it has not yet been resumed. When `can_resume()` returns `true`, the `std::fiber_handle` instance may be resumed by `resume()`, `resume_with()`, `resume_from_any_thread()` or `resume_from_any_thread_with()`.

```
bool can_resume() noexcept (1)
```

1) returns `true` if the calling thread is the same as the thread on which `*this` was suspended. If the `std::fiber_handle` has not yet run and has therefore never been suspended, returns `true` as well.

Precondition

1) `*this` represents a valid fiber (`operator bool()` returns `true`)

Notes

`can_resume()` is not marked `const` because in at least one implementation, it requires an internal context switch.

`operator bool` test whether `std::fiber_handle` is valid

```
explicit operator bool() const noexcept (1)
```

1) returns `true` if `*this` represents a fiber of execution, `false` otherwise.

Notes

A `std::fiber_handle` instance might not represent a valid fiber for any of a number of reasons.

- It might have been default-constructed.
- It might have been assigned to another instance, or passed into a function.
`std::fiber_handle` instances are move-only.
- It might already have been resumed – calling `resume()`, `resume_with()`, `resume_from_any_thread()` or `resume_from_any_thread_with()` invalidates the instance.
- The *entry-function* might have voluntarily terminated the fiber by returning.

The essential points:

- Regardless of the number of `std::fiber_handle` declarations, exactly one `std::fiber_handle` instance represents each suspended fiber.
- No `std::fiber_handle` instance represents the currently-running fiber.

(comparisons) establish an arbitrary total ordering for `std::fiber_handle` instances

```
bool operator<(const fiber_handle& other) const noexcept (1)
```

1) This comparison establishes an arbitrary total ordering of `std::fiber_handle` instances, for example to store in ordered containers. (However, key lookup is meaningless, since you cannot construct a search key that would compare equal to any valid entry.) There is no significance to the relative order of two instances.

`swap` swaps two `std::fiber_handle` instances

```
void swap(fiber_handle& other) noexcept (1)
```

1) Exchanges the state of `*this` with `other`.

std::unwind_fiber() terminate the current running fiber, switching to the fiber represented by the passed `std::fiber_handle`. This is like returning that `std::fiber_handle` from the *entry-function*, but may be called from any function on that fiber.

```
void unwind_fiber(fiber_handle&& other) (1)
```

- 1) throws `std::unwind_exception`, binding the passed `std::fiber_handle`. The running fiber's first stack entry catches `std::unwind_exception`, extracts the bound `std::fiber_handle` and terminates the current fiber by returning that `std::fiber_handle`.

Parameters

other the `std::fiber_handle` to which to switch once the current fiber has terminated

Preconditions

- 1) `other` must be valid (`operator bool()` returns `true`)

Return value

- 1) None: `std::unwind_fiber()` does not return

Exceptions

- 1) throws `std::unwind_exception`

std::unwind_exception is the exception used to unwind the stack referenced by a `std::fiber_handle` being destroyed. It is thrown by `std::unwind_fiber()`. `std::unwind_exception` binds a `std::fiber_handle` referencing the fiber to which control should be passed once the current fiber is unwound and destroyed.

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